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United States Patent Application Publication

20250282175

Kind Code

A1

Publication Date

September 11, 2025

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FREEWHEEL ASSEMBLY CONVERTIBLE BETWEEN FREEWHEEL AND FIXED-GEAR MODES

Abstract

A freewheel assembly has a hub, a brake mechanism, and at least one switch. The hub has an axle, a hub shell mounted around the axle, a drive screw disposed at one of two ends of the hub shell; and a sprocket mounted around the drive screw and having an inner shell and an outer shell. The inner shell is screwed with the drive screw and has a first engaging notch. The outer shell is mounted around the inner shell and has a second engaging notch. The brake mechanism has a clutch cone configured to be driven by the drive screw to move along the axle and at least one brake shoe disposed to an exterior circumference of the clutch cone. Each switch is screwed to the outer shell and has an engaging block engaging with at least one of the first engaging notch and the second engaging notch.

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Family ID: 1000007746063

Appl. No.: 18/595485

Filed: March 05, 2024

Publication Classification

Int. Cl.: B60B27/04 (20060101); B60B27/00 (20060101); B60B27/02 (20060101); B62L1/02 (20060101); B62L5/08 (20060101)

U.S. Cl.:

CPC B60B27/047 (20130101); B60B27/0047 (20130101); B60B27/023 (20130101); B62L1/02 (20130101); B62L5/08 (20130101); B60B2200/47 (20130101);

Background/Summary

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0001] The present invention relates to a drivetrain of a bicycle, and more particularly to a freewheel assembly that can be switched between a freewheel mode and a fixed-gear mode.

2. Description of Related Art

[0002] A bicycle, also known as a bike or a cycle, is a common small land vehicle with two wheels. A rider of the bicycle can stamp pedals of the bicycle to power the bicycle. Therefore, the bicycle is considered as a green and environmentally friendly means of transportation. Manufacture of bicycles is a traditional industry with a history of more than 100 years. The focus of the world's bicycle industry is changing from the traditional means of transportation to sport bicycles, mountain bicycles, and leisure bicycles.

[0003] There are many common types of bicycles, such as single speed bicycles with coaster brakes, fixed-gear bicycles, and single speed bicycles. In a single speed bicycle with a coaster brake, pedals and a sprocket of the bicycle cannot backward rotate together, and such bicycle has to brake by backward stamping the pedals half circle. In a fixed-gear bicycle, pedals and the sprocket of the bicycle are always backward and forward rotate with each other, and such bicycle has to brake by backward stamping the pedals. Finally, a drivetrain of a single speed bicycle provides only forward movement. Various brake functions of bicycles are achieved by different hubs. However, the hub is hard to be altered, and a rider has to purchase different kinds of bicycles to experience the various brake functions. Therefore, it is urgent to design a freewheel assembly with convertible functions to solve the above problems.

[0004] To overcome the shortcomings of the conventional hubs, the present invention provides a freewheel assembly to mitigate or obviate the aforementioned problems.

SUMMARY OF THE INVENTION

[0005] The main objective of the present invention is to provide a freewheel assembly that can convert to a fixed-gear mode.

[0006] The freewheel assembly has a hub, a brake mechanism, and at least one switch. The hub has an axle, a hub shell mounted around the axle, a drive screw disposed at one of two ends of the hub shell; and a sprocket mounted around the drive screw and having an inner shell and an outer shell. The inner shell is screwed with the drive screw and has a first engaging notch. The outer shell is mounted around the inner shell and has a second engaging notch. The brake mechanism has a clutch cone configured to be driven by the drive screw to move along the axle and at least one brake shoe disposed to an exterior circumference of the clutch cone. Each switch is screwed to the outer shell and has an engaging block engaging with at least one of the first engaging notch and the second engaging notch.

[0007] Other objects, advantages, and novel features of the invention will become more apparent from the following detailed description when taken in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 is a perspective view of a first embodiment of a freewheel assembly in accordance with the present invention;

[0009] FIG. 2 is a perspective view of the freewheel assembly in FIG. 1, shown with a hub shell of the freewheel assembly omitted;

[0010] FIG. 3 is a perspective view of a second embodiment of the freewheel assembly in accordance with the present invention;

[0011] FIG. 4 is a perspective view of the second embodiment of the freewheel assembly in FIG. 3, shown with a fixing plate of the freewheel assembly omitted;

[0012] FIG. 5 is a perspective view of a brake cone of a brake mechanism of the freewheel assembly of the present invention;

[0013] FIG. 6 is a perspective view of a brake shoe of the brake mechanism of the freewheel assembly of the present invention;

[0014] FIG. 7 is a side view in partial section of the freewheel assembly in FIG. 1;

[0015] FIG. 8 is a cross-sectional side view of the freewheel assembly in FIGS. 1; and

[0016] FIG. 9 is a cross-sectional side view of the freewheel assembly in FIG. 3.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

[0017] With reference to FIGS. 1, 2, 7, and 8, a first embodiment of a freewheel assembly in accordance with the present invention has a hub **100**, a brake mechanism **200**, at least one switch **300**, and a spring holder **400**. The hub **100** has an axle **101**, a hub shell **102**, a drive screw **103**, a brake arm **104**, a sprocket **105**, a lock nut **106**, and three ball bearing retainers **107**. The hub shell **102** is mounted around the axle **101** and has two opposite ends. The drive screw **103** is disposed at one of the two ends of the hub shell **102**. The brake arm **104** is disposed at the other one of the two ends of the hub shell **102**. The sprocket **105** is screwed to an exterior circumference of the drive screw **103** and is more steadily than a connection provided by splines of bike cassettes. The sprocket **105** has an inner shell **1051** and an outer shell **1052**. The inner shell **1051** is screwed with the drive screw **103** and engages with the outer shell **1052** by a ratchet. The inner shell **1051** has at least one first engaging notch **1053** disposed at an exterior rim of the inner shell **1051**. The outer shell **1052** has at least one second engaging notch **1054** disposed at an exterior rim of the outer shell **1052**. The lock nut **106** is disposed aside the sprocket **105**.

[0018] The brake mechanism **200** is disposed within the hub shell **102** and has a clutch cone **201**, a brake cone **202**, and at least one brake shoe **203**. The clutch cone **201** is mounted around the axle **101** and is screwed with the drive screw **103**. The brake cone **202** is mounted around and fixed to the axle **101**. The spring holder **400** is disposed at an end, of the brake cone **202**, heading to the clutch cone **201**. The spring holder **400** has at least one limiting strip **401** fixed to an end of the spring holder **400**. The brake cone **202** has at least one limiting groove **2021** and disposed at an end, of the brake cone **202**, heading to the clutch cone **201**. The at least one limiting groove **2021** is configured to respectively accommodate the at least one limiting strip **401**. Each of the at least one brake shoe **203** is disposed to an exterior circumference of the brake cone **202** and the exterior circumference of the clutch cone **201**.

[0019] Each of the at least one switch **300** is fixed above a corresponding one of the at least one second engaging notch **1054**. In addition, each switch **300** has a screwing plate **301**, an adjusting bolt **302**, an adjusting nut **303**, and an engaging block **304**. The screwing plate **301** has two opposite ends screwed to the outer shell **1052**. The adjusting bolt **302** passes through the screwing plate **301**. The adjusting nut **303** is threaded with the adjusting bolt **302** and is disposed at a top of the screwing plate **301**. The engaging block **304** is threaded with the adjusting bolt **302** and is disposed below the screwing plate **301**. As the adjusting nut **303** is rotated, the adjusting bolt **302** moves axially to adjust a position of the engaging block **304** along the adjusting bolt **302**. The freewheel assembly of the present invention is in a freewheel mode as the engaging block **304** is disposed in the corresponding one of the at least one second engaging notch **1054**. When a rider pedals backward to rotate the sprocket **105**, the drive screw **103** is still, and the freewheel assembly performs as a normal freewheel. When the engaging block **304** is disposed in a corresponding one of the at least first engaging notch **1053** below the abovementioned second engaging notch **1054**,

the freewheel assembly of the present is switched to a fixed-gear mode. When the rider pedals backward to rotate the sprocket **105**, the drive screw **103** is driven by the sprocket **105** to push the clutch cone **201** till said limiting strip **401** of the spring holder **400** is inserted into said limiting groove **2021** of the brake cone **202**. The limiting strip **401** is blocked by the limiting groove **2021**, and the clutch cone **201** cannot rotate backward. Meanwhile, the clutch cone **201** pushes said brake shoe **203**, and the brake shoe **203** is pushed to rub an interior circumference of the hub shell **102** to achieve a brake function. The above-mentioned brake function is conventional and is not further described in detail. Practically, a compression spring may be deployed between the screwing plate **301** and the engaging block **304** to push the engaging block **304**.

[0020] In addition, the brake cone **202** has at least one positioning notch **2023** disposed at the exterior circumference of the brake cone **202**. Each brake shoe **203** has a positioning protrusion **2031** engaging with a corresponding one of the at least one positioning notch **2023** of the brake cone **202**. Each positioning notch **2023** has a first curved face therein. Each brake shoe **203** has a second curved face disposed at a side of the brake shoe **203**. The second curved face of each brake shoe **203** is configured to fit the first curved face of the corresponding positioning notch **2023** of the brake cone **202**. Consequently, each brake shoe **203** can be positioned by said positioning notch **2023**, and a gap between the brake shoe **2023** and the brake cone **202** is reduced.

[0021] Specifically, said at least one first engaging notch **1053** is implemented as two first engaging notches **1053**, and said at least one second engaging notch **1054** is implemented as two second engaging notches **1054**. The two first engaging notches **1053** and the two second engaging notches **1054** are aligned with one another. Therefore, position of the engaging block **304** of each switch **300** can be easily adjusted inside a corresponding one of the first engaging notches **1053** and a corresponding one of the second engaging notches **1054** to switch the freewheel assembly of the present invention between the freewheel mode and the fixed-gear mode.

[0022] Wherein, an opening of each first engaging notch **1053** and an opening of each second engaging notch **1054** are in the same size and are both larger than a cross-sectional area of the engaging block **304** of each switch **300**. Consequently, the engaging block **304** can be embedded in one corresponding first engaging notch **1053** and in one corresponding second engaging notch **1054** certainly. Furthermore, a height of the engaging block **304** of each switch **300** is equal to a depth of each second engaging notch **1054** to prevent movement of the outer shell **1052** from interfering with movement of the inner shell **1051**.

[0023] Specifically, the three ball bearing retainers **107** of the hub **100** are respectively disposed at the exterior circumference of the brake cone **202**, a middle of the exterior circumference of the drive screw **103**, and an end of an interior circumference of the drive screw **103**. The three ball bearing retainers **107** support the brake cone **202** and the drive screw **103** and promote stability of the structure of the freewheel assembly of the present invention. The three ball bearing retainers **107** also reduce the friction produced during operation of the freewheel assembly of the present invention and make said freewheel easy to rotate.

[0024] A second embodiment:

[0025] With reference to FIGS. **3**, **6**, and **9**, the second embodiment differs from the first embodiment in that each switch **300** has a fixing plate **305** and an engaging block **306**. The fixing plate **305** has two opposite ends screwed to the outer shell **1052**. The engaging block **306** is disposed at a bottom of the fixing plate **305**. In the second embodiment, the freewheel assembly of the present invention is switched from the freewheel mode to the fixed-gear mode by the engaging block **306** engaging in one corresponding second notch **1054** and one corresponding first engaging notch **1053** below said corresponding second engaging notch **1054**. When the rider stamps backward to rotate the sprocket **105**, the drive screw **103** is driven by the sprocket **105** to push the clutch cone **201** till said limiting strip **401** of the spring holder **400** is inserted into said limiting groove **2021** of the brake cone **202**. The limiting strip **401** is blocked by the limiting groove **2021**, and the clutch cone **201** cannot rotate backward. Meanwhile, the clutch cone **201** pushes said brake

shoe **203**, and the brake shoe **203** is pushed to rub an interior circumference of the hub shell **102** to achieve a brake function. When the fixing plate **305** and the engaging block **306** are removed, the freewheel assembly of the present invention is switched from the fixed-gear mode to the freewheel mode. When the rider stamps backward to rotate the sprocket **105**, the drive screw **103** is still, and the freewheel assembly performs as a normal freewheel.

[0026] Working principle of the freewheel assembly of the present invention: when the freewheel assembly of the present invention is in use, the sprocket **105** is mounted around the drive screw **103** and fixed by tightening the lock nut **106**. By pushing each engaging block **304/306** of each switch **300** in one corresponding first engaging notch **1053** and one corresponding second engaging notch **1054**, the freewheel assembly of the present invention is switched from the freewheel mode of the fixed-gear mode. When the rider stamps backward to rotate the sprocket **105**, the drive screw **103** is driven by the sprocket **105** to push the clutch cone **201** till said limiting strip **401** of the spring holder **400** is inserted into said limiting groove **2021** of the brake cone **202**. The limiting strip **401** is blocked by the limiting groove **2021**, and the clutch cone **201** cannot rotate backward.

Meanwhile, the clutch cone **201** pushes said brake shoe **203**, and the brake shoe **203** is pushed to rub an interior circumference of the hub shell **102** to achieve a brake function. The freewheel assembly of the present invention is switched back to the freewheel mode to operate as a normal freewheel by disengaging each engaging block **304** from said corresponding first engaging notch **1053** or by removing the fixing plate **305** and the engaging block **306**. The freewheel assembly of the present invention has an advance design and a compact structure, is easy to use, and operates reliably. Conversion between the freewheel mode and the fixed-gear mode is achieved via said first engaging notch **10523** of the inner shell **1051**, said second engaging notch **1054** of the outer shell **1052**, and said switch **300**.

[0027] The freewheel assembly of the present invention meets people's demands for different functions of bicycles and massively reduces the cost of purchasing components of bicycles.

[0028] Even though numerous characteristics and advantages of the present invention have been set forth in the foregoing description, together with details of the structure and features of the invention, the disclosure is illustrative only. Changes may be made in the details, especially in matters of shape, size, and arrangement of parts within the principles of the invention to the full extent indicated by the broad general meaning of the terms in which the appended claims are expressed.

Claims

1. A freewheel assembly comprising: a hub having an axle; a hub shell mounted around the axle and having two opposite ends; a drive screw disposed at one of the two ends of the hub shell; and a brake arm disposed at the other one of the two ends of the hub shell; a sprocket mounted around the drive screw and having an inner shell screwed with an exterior circumference of the drive screw and having a first engaging notch disposed at an exterior rim of the inner shell; and an outer shell mounted around the inner shell and having a second engaging notch disposed at an exterior rim of the outer shell; a brake mechanism having a clutch cone disposed within the hub shell, mounted around the axle, and configured to be driven by the drive screw to move along the axle; a brake cone disposed within the hub shell and fixed to the axle; and at least one brake shoe disposed within the hub shell and disposed to an exterior circumference of the clutch cone and an exterior circumference of the brake cone; and at least one switch, each of the at least one switch screwed to the outer shell of the sprocket and having an engaging block engaging with at least one of the first engaging notch of the inner shell and the second engaging notch of the outer shell.

2. The freewheel assembly as claimed in claim 1, wherein each of the at least one switch has a screwing plate having two ends screwed to the outer shell of the sprocket; an adjusting bolt inserted through the screwing plate; and an adjusting nut threaded with the adjusting bolt and disposed at a

top of the screwing plate; and the engaging block is threaded with the adjusting bolt and is disposed below the screwing plate.

3. The freewheel assembly as claimed in claim 1, wherein each one of the at least one switch has a fixing plate having two ends screwing to the outer shell of the sprocket; and the engaging block is fixed to a bottom of the fixing plate.

4. The freewheel assembly as claimed in claim 1, wherein the brake cone has at least one positioning notch disposed at the exterior circumference of the brake cone; and each one of the at least one brake shoe has a positioning protrusion engaging with a corresponding one of the at least one positioning notch of the brake cone.

5. The freewheel assembly as claimed in claim 1, wherein the at least one first engaging notch of the inner shell includes two said first engaging notches; the at least one second engaging notch of the outer shell includes two said second engaging notches; and the two first engaging notches and the two second engaging notches are aligned.

6. The freewheel assembly as claimed in claim 2, wherein an opening of each one of the at least one first engaging notch and an opening of each one of the at least one second engaging notch are in the same size and are both larger than a cross-sectional area of the engaging block of each one of the at least one switch.

7. The freewheel assembly as claimed in claim 1, wherein the hub has three ball bearing retainers respectively disposed at the exterior circumference of the brake cone, a middle of the exterior circumference of the drive screw, and an end of an interior circumference of the drive screw.

8. The freewheel assembly as claimed in claim 2, wherein a compression spring is deployed between the screwing plate and the engaging block of each one of the at least one switch.

9. The freewheel assembly as claimed in claim 1, wherein the hub has a lock nut disposed aside the sprocket.

10. The freewheel assembly as claimed in claim 2, wherein the hub has a lock nut disposed aside the sprocket.

11. The freewheel assembly as claimed in claim 3, wherein the hub has a lock nut disposed aside the sprocket.

12. The freewheel assembly as claimed in claim 4, wherein the hub has a lock nut disposed aside the sprocket.

13. The freewheel assembly as claimed in claim 5, wherein the hub has a lock nut disposed aside the sprocket.

14. The freewheel assembly as claimed in claim 6, wherein the hub has a lock nut disposed aside the sprocket.

15. The freewheel assembly as claimed in claim 7, wherein the hub has a lock nut disposed aside the sprocket.

16. The freewheel assembly as claimed in claim 1, wherein the freewheel assembly has a spring holder disposed at an end of the brake cone and having at least one limiting strip; and the brake cone has at least one limiting groove configured to respectively accommodate the at least one limiting strip.

17. The freewheel assembly as claimed in claim 2, wherein the freewheel assembly has a spring holder disposed at an end of the brake cone and having at least one limiting strip; and the brake cone has at least one limiting groove configured to respectively accommodate the at least one limiting strip.

18. The freewheel assembly as claimed in claim 3, wherein the freewheel assembly has a spring holder disposed at an end of the brake cone and having at least one limiting strip; and the brake cone has at least one limiting groove configured to respectively accommodate the at least one limiting strip.

19. The freewheel assembly as claimed in claim 4, wherein the freewheel assembly has a spring holder disposed at an end of the brake cone and having at least one limiting strip; and the brake

cone has at least one limiting groove configured to respectively accommodate the at least one limiting strip.

20. The freewheel assembly as claimed in claim 5, wherein the freewheel assembly has a spring holder disposed at an end of the brake cone and having at least one limiting strip; and the brake cone has at least one limiting groove configured to respectively accommodate the at least one limiting strip.
